Verification guide

Rank-six categorification through multiplicity nine

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Research status. AI-generated research · independent verification pending. Treat this manuscript and its accompanying research data with caution: they have not been independently verified by a human mathematician or formalized in Lean.

1 Scope and authoritative records

The package classifies categorifiability of the 25421 rank-six fusion rings of multiplicity at most nine in the published *Fusion Ring Census*. The current ledger has 25362 general exclusions, 56 unitary-positive rings, one spherical-positive ring admitting no pseudo-unitary categorification, and 2 unresolved rings. Positive based rings, rather than the number of constructions, are counted: the 57 positive rings have 64 constructor records.

The authoritative files are `data/census.json`, `data/decisions.json` and `data/classification_counts.json`. The CSV ledger and paper tables are derived from them. The model catalogue records the source, categorical structure and complete basis permutation for each model. The file `data/unresolved_tensors.json` contains all complete remaining tensors. General, pivotal, spherical, pseudo-unitary and unitary existence are separate fields.

The census is pinned to *Fusion Ring Census*, revision

`2a340871242879665fa7ba2b01fa15b8bdeaada3`.

Its rank-six exact-multiplicity counts are $39, 154, 384, 872, 1582, 2768, 3814, 6999, 8809$. The compressed tensor file `data/rank6_tables.txt.gz` has SHA-256

`0e7552e17620452f237e94751c40cd93b0757a466168629db8cc1bf0b9451d00`.

The source and both compressed and uncompressed hashes are recorded in `provenance/repository_source.json`. Import checks fix the input correspondence; completeness of the counting input is supplied by the cited census. The classification does not rerun census enumeration.

2 Verification scope and reproducible commands

The completed verification campaign covers all 25 362 general-exclusion certificates using the unchanged mathematical checker. Its full initial invocation records 25 355 passes and seven failures, with exit status one. Complete replays of all seven failed certificates then pass with SymPy's Python ground types selected before import. The two post-loop count and positive-registry checks also pass when run separately. The initial invocation completed the census import, positive-model and pre-negative checks. The aggregate receipt binds every accepted certificate to its source and input hashes and checks complete coverage.

The seven selected certificates are $R^{(6)}_{4,416}$, $R^{(6)}_{4,427}$, $R^{(6)}_{4,446}$, $R^{(6)}_{5,638}$, $R^{(6)}_{6,1035}$, $R^{(6)}_{8,311}$ and $R^{(6)}_{8,3664}$. Their saved separating vectors use a positional ordering of exact balance equations. Different

string representations of algebraic coefficients in SymPy's Python and Flint backends can change that ordering and its row signs. The diagnostic packets verify the corresponding exact signed permutations. Neither the mathematical checker nor the certificate vectors were relaxed or replaced.

This is a combined campaign, not a single successful full invocation or an all-Python full-census replay. The initial invocation did not record its loaded backend and version at launch; the selected replays record theirs explicitly. The complete scope and preserved failures are in `evidence/prepublication_20261017/rank6_backend/aggregate_summary.json`. The accompanying coverage file contains one record for every excluded ring. These checks do not provide independent human verification or a formal proof in Lean.

After restoring the repository payloads and installing the dependencies in `supplements/rank6/requirements.txt`, the supplied orchestration entry point is, from the repository root:

```
python evidence/prepublication_20261017/rank6_backend/replay_complete.py \
  --repo . --output-dir /tmp/rank6-fresh-campaign --jobs 2
```

The output directory must not already exist. C++17 is required for the finite-search helpers. The orchestrator records actual process backends, preserves failures, repeats every failed certificate with Python ground types, runs the post-loop checks and verifies complete coverage. Its identity, coverage and rejection controls have been tested; the orchestrator itself has not been executed end to end. The completed campaign used the separately recorded full invocation and selected replay commands.

Individual package checks, from `supplements/rank6/`, include

```
python check_manifest.py
python software/current_regressions.py
python -O software/conformal_models.py --verify
python -O software/replay_m9.py --jobs 2
python -O software/replay_m9_structure.py
python -O software/replay_m9_completion.py
python software/release_consistency.py
```

The manifest should be checked before generating new report files. The optional `--proofs-only` mode of `verify.py` omits positive reconstruction and is not a complete release check. The structure-only replay concerns its specified categorical structures; it cannot establish unrestricted nonexistence.

A negative checker reconstructs number fields, finite branch domains, signed quotients, separating vectors and exact inequalities from the full tensor. Numerical optimizers may propose certificates during discovery; their entry points are disabled during certificate checking. An incomplete search, timeout or numerical infeasibility flag cannot establish an exclusion. A successful necessary system cannot establish existence. Normal and optimized runs of the continuation checks verify that their correctness checks do not depend on removable assertions. The categorical theorems and the proofs of their application remain mathematical inputs. Checksums establish file integrity, not a mathematical conclusion.

The multiplicity-five completion. The exclusion of $R^{(6)}_{5,1204}$ first reconstructs its complete 171-profile domain. A forced central product removes 72 profiles; integral cyclic-rotation multiplicities remove the other 99 using 112 certified integer bounds and 99 final exact separating rays. The proof and solver-disabled replay are under `research/exclusions/R6_5_1204/`:

```
python -O research/exclusions/R6_5_1204/verify_exclusion.py
```

The inherited finite-profile certificate remains part of the proof.

3 Completion at multiplicity nine

Every one of the 8 809 rings of exact multiplicity nine is generally excluded. The eleven final exclusions combine signed dimension-field and different-ideal restrictions, diagonal associativity and trace saturation, complete Frobenius–Schur indicator systems, and a Galois spin-orbit bound. The proof does not assume the pivotality conjecture.

The certificate ledger is `certificates/m9_completion.jsonl`. Portable proof bundles are in `research/exclusions/m9_completion/`; each contains its complete input tensors, mathematical argument and exact finite replay. Run

```
python software/replay_m9_completion.py
python -O software/replay_m9_completion.py
```

The source-bound receipt is recorded in `verification/m9/completion/`. The checker regenerates the required finite domains, verifies every branch and rejects corrupted certificates. All correctness checks remain active with `-O`. This calculation concerns categorification; its counting input is the cited census.

4 The three conformal constructions

The positive registry `data/conformal_positive_models.json` adds $R^{(6)}_{4,22}$, $R^{(6)}_{4,91}$ and $R^{(6)}_{4,629}$. Their models are the neutral components of the module categories for $(B_2)_7 \subset (D_7)_1$, $(C_3)_4 \subset (B_{10})_1$ and $(B_2)_{12} \subset (E_8)_1$, respectively. The source files, exact certificates and independent mathematical audit are organized under `research/structural_models/`.

Each conformal embedding supplies an actual unitary algebra and module category. The parity grading gives a rank-six neutral component. Published action graphs or exact free-induction Gram matrices determine a small set of possible fusion tensors. Local modules, duality, formal-codegree induction and the known center's dimensions or indicators distinguish the required tensor. Every resulting tensor is compared in all 216 entries with the census representative. The proofs do not infer a category from dimensions, a modular invariant, or matching modular data.

The canonical command

```
python -O software/conformal_models.py --verify
```

checks the source bindings and exact tensor identifications. The per-model directories `quartic/`, `c3_model/` and `r629/` contain the reconstruction programs and mathematical proofs, including the deeper graph, Weyl-character and indicator checks. These constructions prove existence. They do not classify all associators on the three rings or identify every archived modular-data matrix in a prescribed labelling.

5 The remaining problems

The canonical index `research/frontier/` describes the two remaining general existence questions, their necessary reductions and exact replay commands. No multiplicity-nine case remains open.

The two residuals through multiplicity eight satisfy the following necessary conditions. For $R^{(6)}_{4,155}$, the generalized near-group fusion rules give a uniform sphericality and Galois-pseudo-unitarity proof by one norm argument. Fermion fixed-point traces and odd Gauss sums leave one necessary induction profile, of center rank 52, whose twist spectrum is forced and has exponent 24. The short exact replay is `research/frontier/R6_4_155/verify_center_cuts.py`. Its pointed center also rules out every braided representative of the Morita class. The exact

reconstruction determines every entry of the rank-26 centralizer's modular datum. Its Verlinde coefficients are nonnegative integers, with maximum 9. It forces ten self-dual objects in that factor and twenty in the full center. All stated finite ordinary and generalized indicator checks and all ordered rank-26 Vafa equations pass. No categorical construction follows.

For $R^{(6)}_{8,1660}$, the Ising subcategory makes its complement a regular bimodule. Its relative center has rank eighteen and rules $B[t]/(t^2 - At - L)$, where B is the Ising-center ring. Formal-codegree divisibility and dimension-field constraints give 58 labelled choices. A central second-indicator obstruction and a Morita-invariant invertible count leave one based ring, up to exchanging the Ising factors. The proof and exact replay are in `research/frontier/R6_8_1660/relative_center/`. The retained full-center certificates force exponent dividing 48. Fermion orbit constraints and a weighted second-indicator congruence reduce their 1764 labelled profiles to 1727. The tensor involution and homogeneous-packet Vafa equations then leave 1670; the exact replays are in `research/frontier/R6_8_1660/fermion_profiles/` and `research/frontier/R6_8_1660/spin_profiles/`. A subsequent signed Galois–fermion four-divisibility condition removes raw profile 133 in low frame zero, leaving 1669 profiles in branches $163, 753, 753$. Independently, every indecomposable module category has six simples; full-center saturation, integral Morita Gram reconstruction and a weighted second-indicator obstruction leave three possible dual based rings, of ranks six, six and eight.

The exact calculations and proof audits are under `research/frontier/new_research/`. From the package root, run

```
python research/frontier/new_research/replay.py
```

The summaries of this finite system are in `data/frontier_research_progress.json`.

5.1 Further necessary conditions

The manuscript includes the graded-packet identities, tensor-quotient obstruction, restrictions on Tannakian subcategories and braided Morita models, and exact scope of the analytic tests. The graded-packet conditions leave $1136 = 120 + 508 + 508$ profiles. All-order indicator identities determine the exact central phase masses; the full phase Gram constraints leave $640 = 68 + 286 + 286$ profiles. Exactly eight spin simples have restriction $2X + 2gX + \sigma + 3T$. For $R^{(6)}_{4,155}$, odd cyclic Clifford factorization is a necessary condition at every odd order; all 1144 tested central-target rotation blocks at orders $3, 5, \ldots, 23$ pass. The all-order path-dimension condition is equivalent to even multiplicity in the stated ring family. Both $R^{(6)}_{4,155}$ and $R^{(6)}_{8,1660}$ remain unresolved.

The source-bound receipts and verifiers for these results are retained at the repository path `evidence/residual_20261002/rank6/`; the current research summary and all-order refinements are recorded at `evidence/rigidity_20261017/rank6/`, with its `research_progress.json`. The exact all-exponent Minkowski couplings are retained under `evidence/rank6_laptop_20261002/all_p_minkowski/`. The source hashes in these records identify the inputs required by each checker.

The manuscript further gives a theoretical proof that every tensor functor from a hypothetical categorification of either residual ring is fully faithful. It proves pseudo-unitarity of dominant images and preservation of canonical pivotal structures, then combines these facts with the quotient obstruction and the complete twist-one restriction table. This strengthens the tensor-quotient restriction without changing either unresolved verdict.

5.2 All-order necessary conditions and exact phase masses

From the repository root, run

```
python evidence/rigidity_20261017/rank6/run_checks.py --repo .
python -O evidence/rigidity_20261017/rank6/run_checks.py --repo .
```

The two case folders contain their source-bound replay programs and independent audit receipts. The rank-six $R6_4_155$ checker verifies all 1144 odd cyclic Clifford decompositions and the all-order path criterion; the latter does not exclude multiplicity four. The $R6_8_1660$ checker reconstructs every indicator at orders 1 through 48, all 288 Fourier masses for each conjugate choice, the exact spin restrictions and the 640-profile phase Gram domain. Independent SymPy arithmetic checks the Fourier formulas and the full Ising sequence; a separate coefficient-convolution implementation exhausts the phase allocation domain. Explicit rejection controls and both Python execution modes are included. These checks certify finite arithmetic and necessary conditions; neither residual has a categorification or a general nonexistence proof.

None of these necessary alternatives is a categorical model. A claimed future exclusion must cover the entire necessary domain; a claimed model must have a categorical construction and complete tensor identification.

6 Rebuilding the presentation

After checking the retained evidence, regenerate the ledger exports, model catalogue, tables and PDFs with

```
python software/release.py
python software/model_catalog.py
python software/build_paper_tables.py
python software/build_readme.py
python build_papers.py
```

Run discovery or regeneration in a working copy, because these commands write files. The build publishes PDFs only after successful compilation. All counts must agree with the decision ledger. The supplement retains accepted proofs, needed dependencies and current residual evidence; obsolete searches for the three conformally realized rings are omitted.

7 Fermion linking and center indicators

The manuscript proves the fermion Azumaya linking identity and checks all $22^4 = 234256$ instances on the retained $R6_4_155$ based module. This is a necessary compatibility result, not a construction of module associators. Both rank-six categorification questions remain unresolved. For $R6_8_1660$, an exact character-trace refinement forces $\nu_3(T) = 2$ and leaves two conjugate third-indicator profiles. A separate exact Verlinde calculation excludes the specified rank-37 witness by forcing a fusion coefficient $1/2$; other profiles remain open. The source-bound standard-library replay and proof audit are in `evidence/unified_20261016/rank6/higher_indicators/`; run its `verify_r1660_higher_indicators.py` with `--repo .` from the repository root, also with `-O`. The scoped witness obstruction has its own standard-library checker `verlinde_obstruction/verify_r1660_witness_verlinde.py` under the same evidence directory. It also checks the linear necessary system, which passes. The fermion identity is reproduced from the repository root by

```
python evidence/unified_20261016/rank6/verify_fermion_crossing.py --repo .
```

For the complete continuation, run

```
python3 tools/verify_current_publication.py --historical
```

and repeat with `-O`. This mode reconstructs the source-bound input ledgers required by the certificates before replaying them. It is a continuation check, separate from the full rank-six campaign described above.